

BEE 4750 Lab 2: Monte Carlo

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Due Date

Thursday, 10/02/25, 9:00pm

Setup

The following code should go at the top of most Julia scripts; it will load the local package environment and install any needed packages. You will see this often and shouldn't need to touch it.

```
import Pkg
Pkg.activate(".")
Pkg.instantiate()
```

Activating project at `~/lab-2-cpj35-design`

```
using Random # random number generation
using Distributions # probability distributions and interface
using Statistics # basic statistical functions, including mean
using Plots # plotting
```

Overview

In this lab, we will conduct a Monte Carlo experiment and explore how sensitive Monte Carlo estimates are to the underlying probability distribution(s).

You should always start any computing with random numbers by setting a “seed,” which controls the sequence of numbers which are generated (since these are not *really* random, just “pseudorandom”). In Julia, we do this with the `Random.seed!()` function.

```
Random.seed!(1)
```

`TaskLocalRNG()`

It doesn't matter what seed you set, though different seeds might result in slightly different values. But setting a seed means every time your notebook is run, the answer will be the same.

Seeds and Reproducing Solutions

If you don't re-run your code in the same order or if you re-run the same cell repeatedly, you will not get the same solution. If you're working on a specific problem, you might want to re-use `Random.seed()` near any block of code you want to re-evaluate repeatedly.

Probability Distributions and Julia

Julia provides a common interface for probability distributions with the [Distributions.jl package](#). The basic workflow for sampling from a distribution is:

1. Set up the distribution. The specific syntax depends on the distribution and what parameters are required, but the general call is the similar. For a normal distribution or a uniform distribution, the syntax is

```
# you don't have to name this "normal_distribution"
#  is the mean and  is the standard deviation
normal_distribution = Normal( , )
# a is the upper bound and b is the lower bound; these can be set to +Inf or -Inf for an
uniform_distribution = Uniform(a, b)
```

There are lots of both [univariate](#) and [multivariate](#) distributions, as well as the ability to create your own, but we won't do anything too exotic here.

2. Draw samples. This uses the `rand()` command (which, when used without a distribution, just samples uniformly from the interval $[0, 1]$.) For example, to sample from our normal distribution above:

```
# draw n samples
rand(normal_distribution, n)
```

Putting this together, let's say that we wanted to simulate 100 six-sided dice rolls. We could use a [Discrete Uniform distribution](#).

```
dice_dist = DiscreteUniform(1, 6) # can generate any integer between 1 and 6
dice_rolls = rand(dice_dist, 100) # simulate rolls
```

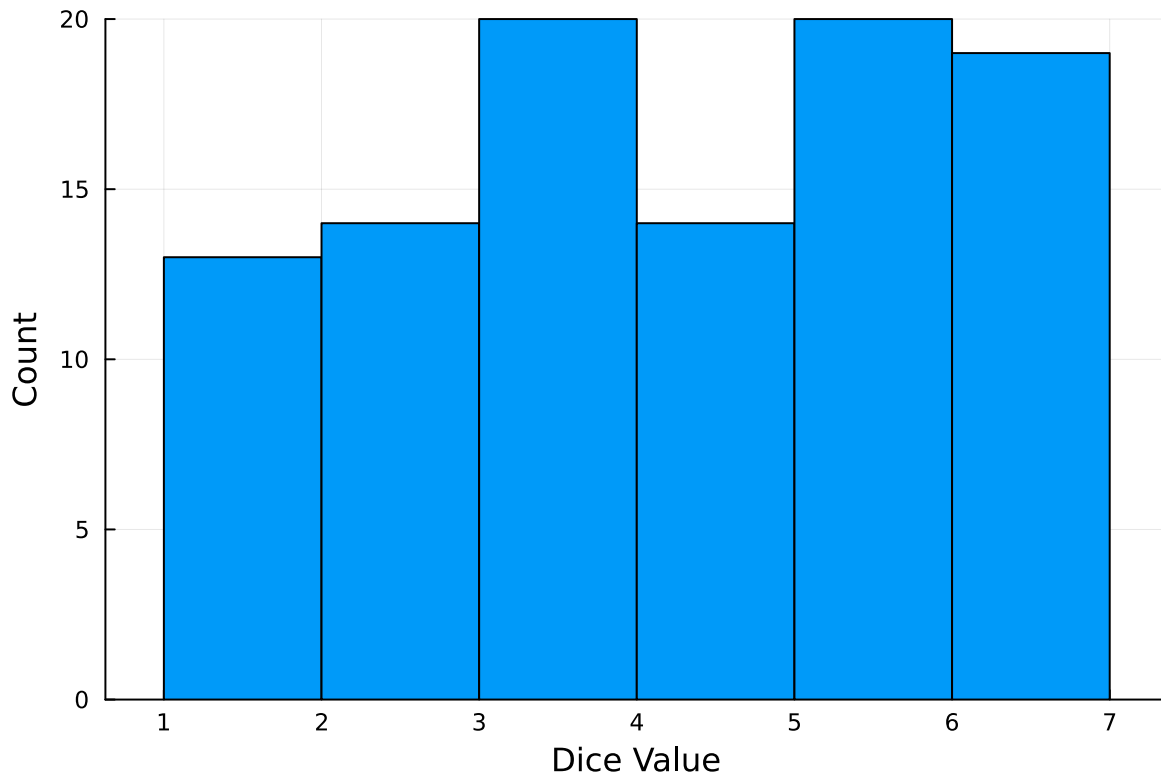
100-element Vector{Int64}:

1
3
5
4
6
2
5
5
5
2
4
3
2

2
3
3
3
6
5
5
6
3
6
6
6

And then we can plot a histogram of these rolls:

```
histogram(dice_rolls, legend=:false, bins=6)
ylabel!("Count")
xlabel!("Dice Value")
```



Instructions

Remember to:

- Evaluate all of your code cells, in order (using a Run All command). This will make sure all output is visible and that the code cells were evaluated in the correct order.
- Tag each of the problems when you submit to Gradescope; a 10% penalty will be deducted if this is not done.

Problem

A common engineering problem is to quantify flood risk, which is typically computed by propagating a flood hazard distribution through a *depth-damage function* relating flood depths

to economic damages. A reasonable depth-damage function for a house without a basement is a bounded logistic function,

$$d(h) = \mathbb{1}_{h>0} \frac{L}{1 + \exp(-k(h - h_0))},$$

where d is the damage as a percent of total structure value, h is the water depth (relative to the house's ground floor elevation) in m, $\mathbb{1}_{x>0}$ is the indicator function, L is the maximum loss in USD, k is the slope of the depth-damage relationship, and h_0 is the inflection point. We'll assume $L = \$200,000$, $k = 0.8$, and $h_0 = 3$.

For this problem, suppose that we have two different probability distributions characterizing annual maxima flood depths:

1. $h \sim \text{LogNormal}(1.2, 0.3)$;
2. $h \sim \text{GeneralizedExtremeValue}(3, 0.9, -0.15)$.

Plot histograms of samples from these distributions and conduct a Monte Carlo experiment for annual maximum flood damages using each. Answer the following questions:

- What are the Monte Carlo estimates of the expected value and the 99% quantile for the annual maximum damage the structure would suffer for each of these flood hazard distributions?
- How did you decide on your sample size?
- Why do you think the estimates differed or did not differ (you can also plot the depth-damage function to compare with the samples)?
- How might you decide (based on getting additional information, if needed) which distribution is “better”?

```
L=200_000
k=0.8
h0=3
sample_size=500_000

damages(h)=L/(1+exp(-k*(h-h0)))

depth_logn = rand(LogNormal(1.2,0.3),sample_size)
depth_GEV = rand(GeneralizedExtremeValue(3,0.9,-0.15),sample_size)

damage_vals_logn = damages.(depth_logn)
damage_vals_GEV = damages.(depth_GEV)

sample_logn_plot = histogram(depth_logn, legend=false, bins=50, title="Log Normal Flood Depth")
display(sample_logn_plot)
```

```

sample_GEV_plot = histogram(depth_GEV, legend=false, bins=50, title="Generalize Extreme Value", xlab="Depth", ylab="Frequency", col="red", border="black")
display(sample_GEV_plot)

log_plot = histogram(damage_vals_logn, legend=false, bins=50, title="Log Normal Damage", xlab="Damage", ylab="Frequency", col="blue", border="black")
display(log_plot)

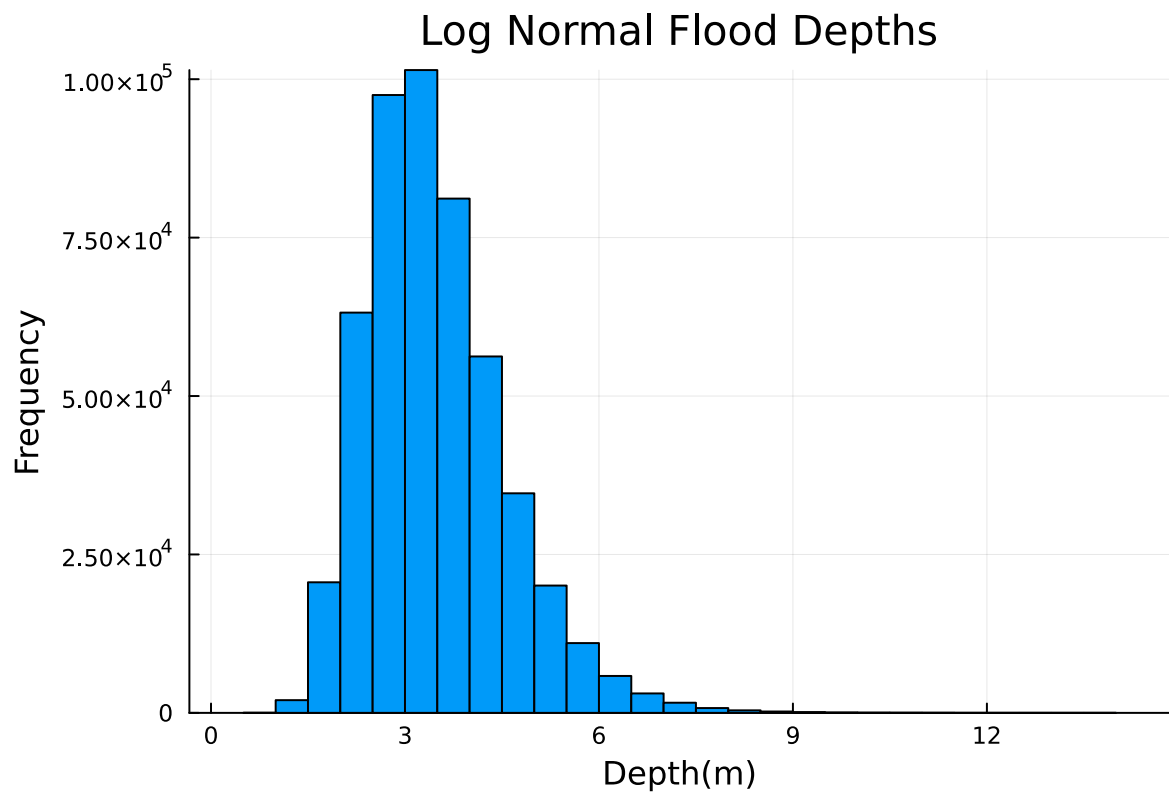
monte_estimate_logn=mean(damage_vals_logn)
quant_99_logn = quantile(damage_vals_logn, 0.99)
println("Average Damages: \$$ (round(monte_estimate_logn,digits=2))\n99% Quantile: \$$ (round(quant_99_logn,digits=2))")
println(error_logn)

GEV_plot = histogram(damage_vals_GEV, legend=false, bins=50, title="Generalized Extreme Value", xlab="Damage", ylab="Frequency", col="red", border="black")
display(GEV_plot)

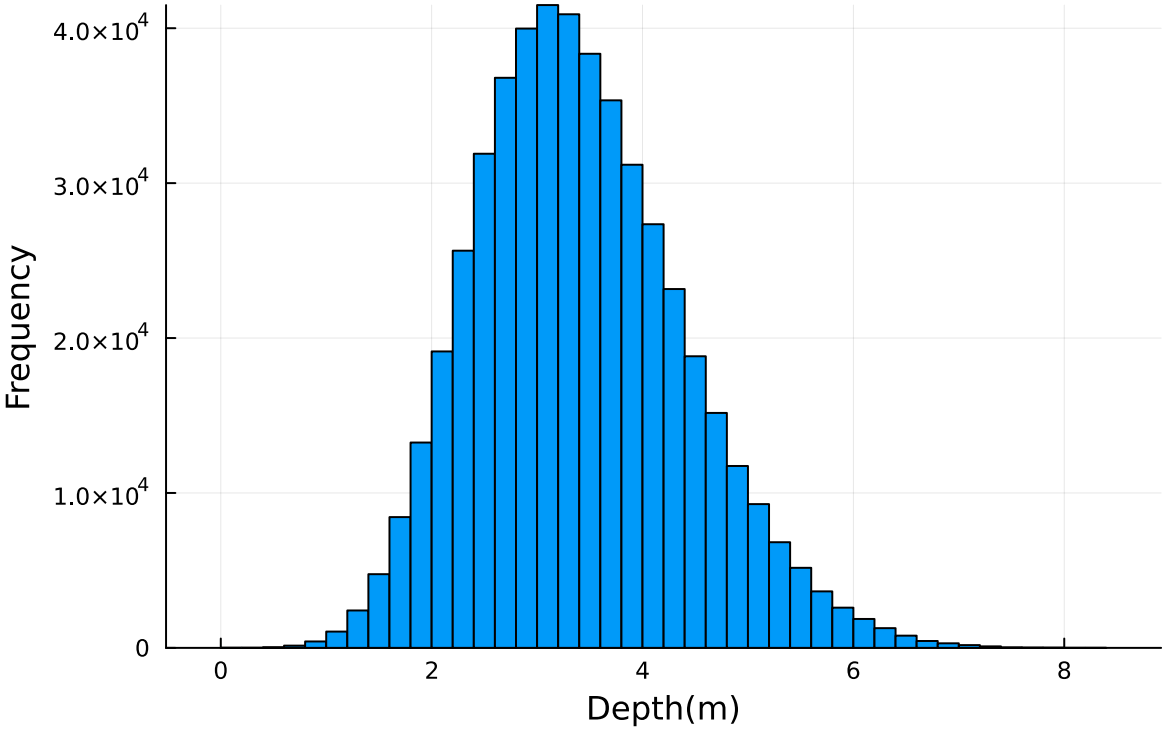
monte_estimate_GEV=mean(damage_vals_GEV)
quant_99_GEV = quantile(damage_vals_GEV, 0.99)
println("Average Damages: \$$ (round(monte_estimate_GEV,digits=2))\n99% Quantile: \$$ (round(quant_99_GEV,digits=2))")

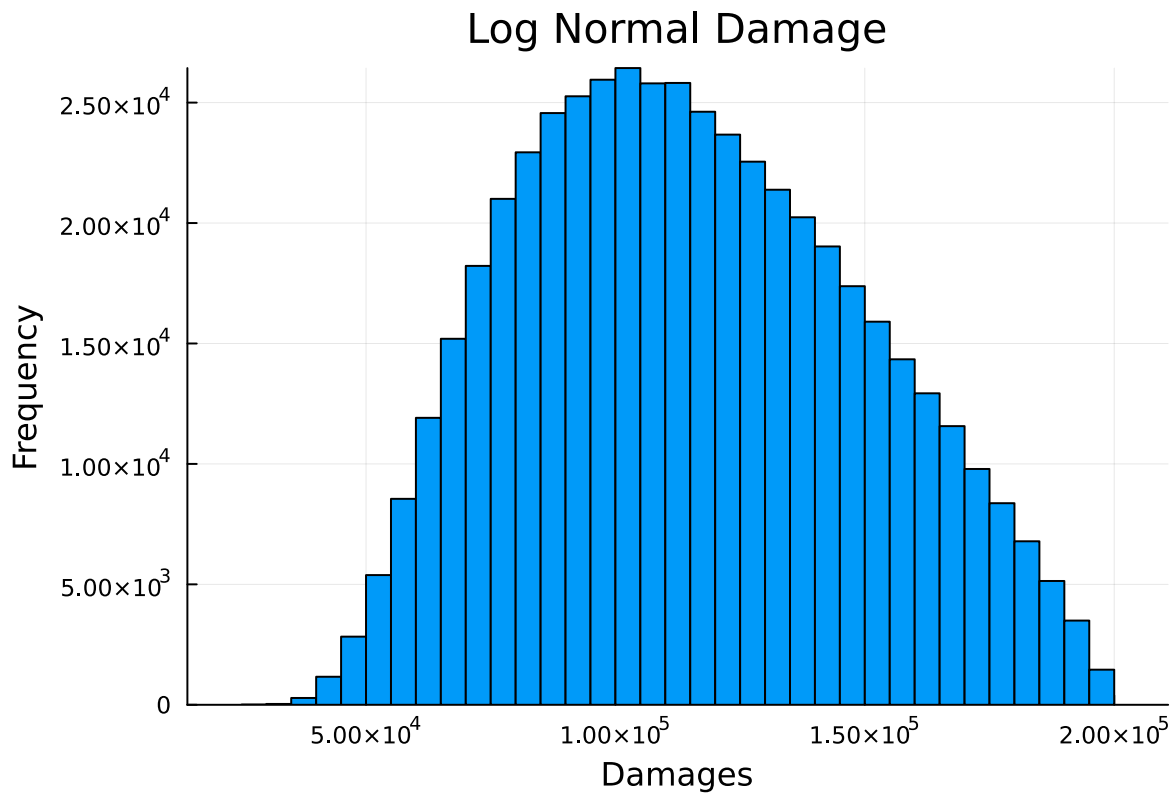
println("I decided on a sample size of 500,000 because after running the code a few times I noticed that the Log Normal plot was more right-skewed than the GEV plot which explains the higher average damage.")
println("The Log Normal plot is more right-skewed than the GEV plot which explains the higher average damage.")
println("In order to determine which strategy was better I would look into previous flood history and see if there are any patterns in the data.")

```

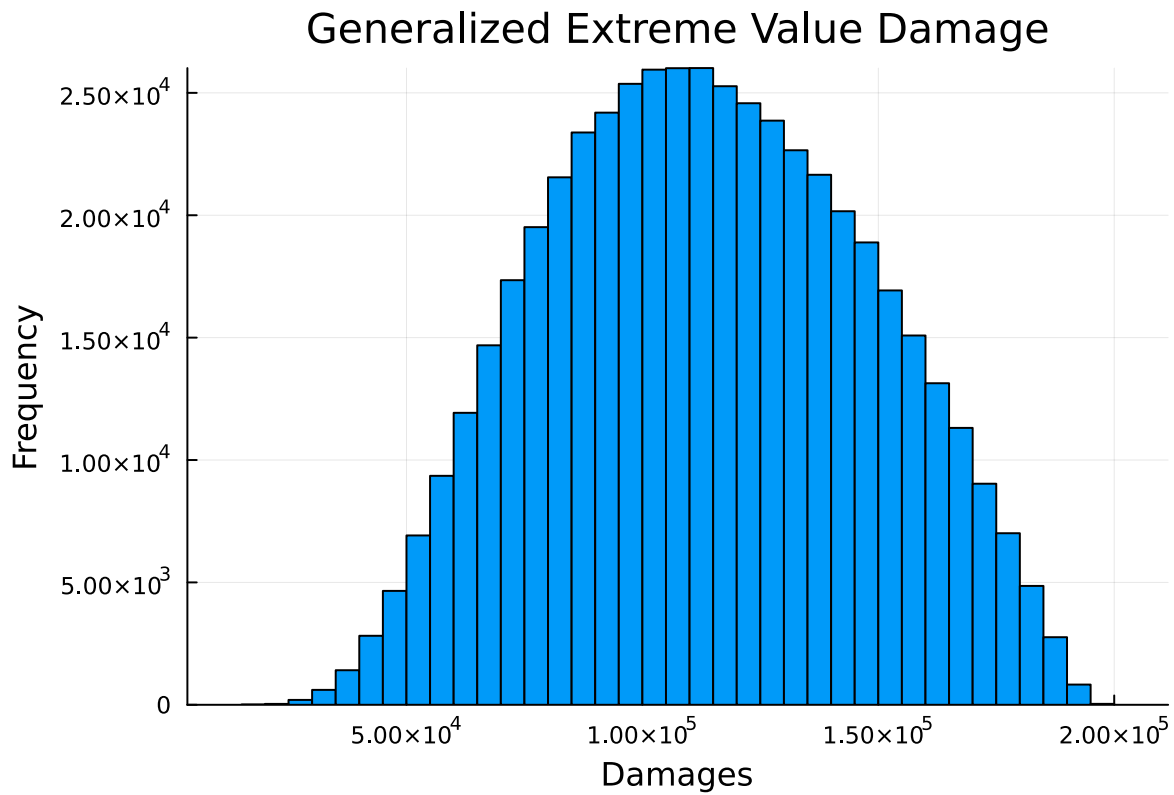


Generalize Extreme Value Flood Depths





Average Damages: \$115211.19
99% Quantile: \$189947.55
34080.837781581045



Average Damages: \$113532.69

99% Quantile: \$183367.4

I decided on a sample size of 500,000 because after running the code a few times I noticed that the

The Log Normal plot is more right-skewed than the GEV plot which explains the higher means of the

In order to determine which strategy was better I would look into previous flood history data and

References

Put any consulted sources here, including classmates you worked with/who helped you.